Name of the Student..Anubhav Ranjan..... Scholar Number..2521201110

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Examination: Mid-Term **Month and Year: October, 2025**

Course: M. Tech 1st Sem. **Branch:** CN & IS **Subject Code:** CSCN-1101; CSIS-1102
Subject Name: Computer & Network Security **Maximum Marks:** 20
Duration: 60 Min. **Date: 29/10/2025** **Time:** 9:30 AM to 10:30 AM

Important Instructions:
1. All questions are compulsory.
2. Do not write anything on the question paper except the entries specifically given (e.g., Name, Scholar No.).
3. Electronic gadgets are not permitted during the exam.
4. Your exam may be canceled if any unfair means are detected during the exam.

Q. No.	Questions	Marks	COs
1.	In a Network Security, data is transmitted over an insecure public communication channel. Explain the mechanisms used to ensure the Confidentiality, Integrity, and Non-repudiation properties of the transmitted data. Illustrate each property with the help of an appropriate figure.	5	CO1
2.	Alice intends to send the message "DEFEND THE WALL" to Bob using the Playfair cipher with the key "NETWORK". Determine the resulting encrypted message, showing all the steps involved in the encryption process. (**NOTE:** Use 'X' as the padding character if required)	2.5	CO2
3.	Explain the reason for selecting the RSA algorithm for secure message transmission over an insecure communication channel, ensuring that the message remains protected from unauthorized access. Also, describe in detail all the steps involved in the RSA algorithm for both the encryption and decryption processes.	2.5	CO2
4..	Employee-1 wants to send a large amount of data to Employee-2 over a network. Since the data is large and must be transmitted securely, it cannot be transmitted directly without using a data compression or transformation technique such as a hash function. To ensure security and enable data verification, Employee-1 applies a hash function before transmission. Analyze this scenario and explain the key security properties of the hash function that are ensured during the transmission process.	5	CO2
5.	Two organizations use the Digital Signature Standard/Algorithm (DSS/DSA) to sign and verify messages. Explain step-by-step the complete DSS digital signature generation and verification process. Also, prove mathematically that, for a valid signature, the verification equation $v \equiv r \pmod q$ always holds true, where v is the value computed by the verifier during verification and r is the first component of the signature produced by the signer.	5	CO3

**********ALL THE BEST**********

Name of the Student..... Scholar Number.2521241110.....

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Examination: End-Term **Month and Year: December, 2025**

Course: M. Tech 1st Sem & Ph.D. **Branch:** CN & IS
Subject Code: CSCN-1101; CSIS-1102
Subject Name: Computer & Network Security **Maximum Marks:** 40
Duration: 120 Minutes **Date:** 09/12/2025 **Time:** 2:30 PM to 4:30 AM

Important Instructions:
1. All questions are compulsory.
2. Do not write anything on the question paper except the entries specifically given (e.g., Name, Scholar No.).
3. Electronic gadgets are not permitted during the exam.
4. Your exam may be canceled if any unfair means are detected during the exam.

Q. No.	Questions	Marks	COs
1.	**A.** Alice wants to send data to Bob over a network that provides basic security services. Explain the network security services typically offered and illustrate them with a suitable diagram showing the flow of secure communication between Alice and Bob.	4	CO1, CO2
	B. In an RSA system with the public parameters (n = 143, e = 7). Determine the following: i. The prime factors ii. Calculate the Euler Totient function φ(n) iii. Find the private key d iv. Decrypt the ciphertext C = 106	4	CO2, CO3
2.	**A.** Explain all the steps involved in the functioning of an Intrusion Detection System (IDS) and an Intrusion Prevention System (IPS). Additionally, draw and label a separate diagram for each mechanism.	4	CO1
	B. In network security, before any communication can take place, a secure session or shared key must be established. Suppose the Diffie-Hellman key exchange protocol is used for this purpose. Explain the steps involved in the key exchange process and illustrate it with a proper transition diagram.	4	CO2, CO3
3.	**A.** Explain the concept of IPSec in providing secure communication over IP networks. Further, draw the packet formats of the Authentication Header (AH) and Encapsulating Security Payload (ESP), clearly labeling each field along with its size.	4	CO1, CO2
	B. A bank's IT security team is investigating suspicious network activity and wants to analyze the efficiency of their firewall in blocking unauthorized access attempts. The bank's firewall is configured to allow only legitimate traffic while blocking malicious or unauthorized requests. The following data is collected over a 5-hour period. Compute the firewall efficiency per hour.	4	CO2

Hour	Total Incoming request	Allowed request	Blocked request
1	500	350	150
2	620	410	210
3	700	500	200
4	800	520	280
5	950	600	350

4.	**A.** A company deploys Kerberos for secure authentication across its enterprise network. Explain the complete Kerberos authentication workflow with the help of suitable diagram.	4	CO2
	B. When a user wants to access services from a service provider, a secure session is established using Secure Sockets Layer (SSL) through the handshake process. Analyze and explain the SSL handshake process with the help of a diagram. Additionally, state whether SSL is considered secure today. If it is, provide a strong reason; if not, mention the improved version that is currently recommended.	4	CO4
5.	**A.** Explain all the components involved in the Wired Equivalent Privacy (WEP) mechanism used in early Wi-Fi networks. Additionally, draw and label the WEP frame format.	5	CO1, CO3
	B. Discuss in short any two of the following: i. Distributed Denial-of-Service (DDoS) attack. ii. Ransomware and Adware iii. Hill Cipher technique iv. DKIM	3	CO1

**********ALL THE BEST**********